

3I/ATLAS Production Rate Table

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Table 1: Catalog of production rates for chemical species Q_X from observations of 3I/ATLAS. Includes error associated with Q_X , observation dates, heliocentric distance r_H , facility used to obtained observation, and relevant citation. Error of zero indicates the production rate is an upper limit and those with three dots indicate a repeated entry. Note that the errors are of the same order of magnitude as the corresponding value of Q_X .

Volatile	Production Rate [s ⁻¹]	Error [s ⁻¹]	Observation Date	r_H [au]	Facility	Reference
H ₂ O	9.10×10^{26}	0.00	07/04/25–05/26/25	4.37	X-SHOOTER	Alvarez-Candal et al. (2025)
	7.40×10^{26}	5.00	07/31/25	3.51	UVOT	Xing et al. (2025)
	3.20×10^{26}	0.20	08/01/25–08/15/25	3.20	SPHEREx	Lisse et al. (2025a)
	2.23×10^{26}	0.08	08/06/25	3.32	JWST/NIRSpec	Cordiner et al. (2025)
	1.50×10^{26}	0.00	08/08/25–08/12/25	3.10–3.30	SPHEREx	Lisse et al. (2025b)
	1.36×10^{27}	0.50	08/19/25	2.90	UVOT	Xing et al. (2025)
	9.46×10^{27}	0.00	08/26/25–09/03/25	2.54	TMRT	Li et al. (2026)
	1.45×10^{28}	0.52	09/08/25–09/09/25	2.27	...	...
	2.08×10^{28}	0.41	09/18/25–09/23/25	1.96	...	...
	8.00×10^{28}	0.80	11/02/25	1.36	MAJIS/JUICE	Bockelee-Morvan et al. (2026)
	1.60×10^{29}	0.00	11/04/25	1.37	ALMA/ACA	Salazar Manzano et al. (2026)
	3.17×10^{29}	0.004	11/06/25	1.40	SOHO/SWAN	Combi et al. (2025)
	2.31×10^{29}	0.003	11/09/25	1.42	...	...
	2.35×10^{29}	0.004	11/10/25	1.43	...	...
	2.36×10^{29}	0.004	11/12/25	1.45	...	...
	6.80×10^{28}	0.70	...	1.45	MAJIS/JUICE	Bockelee-Morvan et al. (2026)
	3.94×10^{28}	1.30	11/13/25	1.46	SOHO/SWAN	Tan et al. (2026)
	2.45×10^{29}	0.004	...	1.47	...	Combi et al. (2025)
	2.09×10^{29}	0.04	11/14/25	1.48	...	...
	5.15×10^{28}	1.65	11/15/25	1.49	...	Tan et al. (2026)
	2.19×10^{29}	0.004	...	1.50	...	Combi et al. (2025)
	4.23×10^{28}	1.37	11/16/25	1.50	...	Tan et al. (2026)
	1.91×10^{29}	0.004	...	1.51	...	Combi et al. (2025)
	3.34×10^{28}	1.07	11/17/25	1.52	...	Tan et al. (2026)
	3.92×10^{28}	1.27	11/18/25	1.53	...	...
	2.51×10^{29}	0.004	...	1.55	...	Combi et al. (2025)
	4.70×10^{28}	0.70	11/19/25	1.56	MAJIS/JUICE	Bockelee-Morvan et al. (2026)
	3.90×10^{29}	0.05	...	1.56	SOHO/SWAN	Combi et al. (2025)
	2.19×10^{29}	0.004	11/20/25	1.58	...	...
	2.55×10^{28}	1.08	11/25/25	1.67	...	Tan et al. (2026)
	4.10×10^{28}	1.20	...	1.68	MAJIS/JUICE	Bockelee-Morvan et al. (2026)
	1.83×10^{28}	0.66	11/26/25	1.69	SOHO/SWAN	Tan et al. (2026)
	1.43×10^{29}	0.004	...	1.68	...	Combi et al. (2025)
	4.08×10^{28}	0.07	12/02/25	1.84	...	...
	2.46×10^{28}	0.76	12/03/25	1.86	...	Tan et al. (2026)
	7.60×10^{27}	2.70	...	1.86	...	Combi et al. (2025)
	1.55×10^{28}	0.66	12/04/25	1.89	...	Tan et al. (2026)
	2.43×10^{28}	0.10	...	1.89	...	Combi et al. (2025)
	1.88×10^{28}	0.66	12/05/25	1.91	...	Tan et al. (2026)
	1.14×10^{28}	0.22	...	1.91	...	Combi et al. (2025)
	1.30×10^{28}	0.45	12/06/25	1.94	...	Tan et al. (2026)
	3.68×10^{28}	0.08	...	1.94	...	Combi et al. (2025)
	1.26×10^{28}	0.51	12/07/25	1.96	...	Tan et al. (2026)
	2.04×10^{28}	0.14	...	1.96	SOHO/SWAN	Combi et al. (2025)
	1.50×10^{28}	0.54	12/08/25	1.99	...	Tan et al. (2026)
	1.40×10^{28}	0.28	12/08/25–12/15/25	2.00–2.20	SPHEREx	Lisse et al. (2026)
	1.00×10^{28}	0.25	12/09/25	1.99	SWAN-SOHO	Combi et al. (2025)
	1.60×10^{28}	0.13	12/10/25	2.01	...	...
	1.13×10^{28}	0.42	12/15/25	2.18	...	Tan et al. (2026)
	1.13×10^{28}	0.48	12/16/25	2.22	SWAN-SOHO	Tan et al. (2026)
	3.78×10^{27}	0.03	12/16/25	2.20	JWST/MIRI	Belyakov et al. (2026)

	1.28×10^{28}	0.47	12/17/25	2.24	SWAN-SOHO	Tan et al. (2026)
	1.61×10^{27}	0.01	12/22/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)
	8.47×10^{26}	0.05	...	2.40	...	Roth et al. (2026a)
	1.57×10^{27}	0.01	...	2.40	...	Roth et al. (2026b)
	9.82×10^{26}	0.13	12/23/25	2.40	...	Roth et al. (2026a)
	1.55×10^{27}	0.12	...	2.40	...	Roth et al. (2026b)
	1.69×10^{27}	0.05	12/23/25	2.40	...	Cordiner et al. (2026)
	1.05×10^{27}	0.02	12/27/25	2.54	JWST/MIRI	Belyakov et al. (2026)
CO ₂	1.70×10^{26}	0.01	08/06/25	3.32	JWST/NIRSpec	Cordiner et al. (2025)
	1.60×10^{27}	0.10	08/01/25–08/15/25	3.20	SPHEREx	Lisse et al. (2025a)
	9.40×10^{26}	0.30	08/08/25–08/12/25	3.10–3.30	SPHEREx	Lisse et al. (2025b)
	8.20×10^{27}	0.80	11/02/25	1.36	MAJIS/JUICE	Bockelee-Morvan et al. (2026)
	6.90×10^{27}	1.00	11/12/25	1.45	...	...
	5.90×10^{27}	1.40	11/19/25	1.56	...	...
	4.00×10^{27}	0.70	11/25/25	1.68	...	...
	3.00×10^{27}	0.45	12/08/25–12/15/25	2.00–2.20	SPHEREx	Lisse et al. (2026)
	8.70×10^{27}	0.09	12/16/25	2.20	JWST/MIRI	Belyakov et al. (2026)
	1.76×10^{27}	0.01	12/23/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)
	1.59×10^{27}	0.01	...	2.40	...	Roth et al. (2026b)
	5.42×10^{27}	0.06	12/27/25	2.54	JWST/MIRI	Belyakov et al. (2026)
CO	1.10×10^{27}	0.00	07/18/25–07/21/25	3.94 – 3.84	JCMT	Hinkle et al. (2025)
	3.70×10^{26}	0.20	08/06/25	3.32	JWST/NIRSpec	Cordiner et al. (2025)
	1.00×10^{26}	0.25	08/01/25–08/15/25	3.20	SPHEREx	Lisse et al. (2025a)
	2.80×10^{26}	0.00	08/08/25–08/12/25	3.10–3.30	...	Lisse et al. (2025b)
	5.75×10^{27}	1.91	09/07/25–09/29/25	2.01	13.7 m MWT	Li et al. (2026)
	6.80×10^{27}	0.11	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
	7.60×10^{27}	1.50	12/08/25–12/15/25	2.00–2.20	SPHEREx	Lisse et al. (2026)
	3.50×10^{27}	0.02	12/23/25	2.40	JWST/NIRSpec	Roth et al. (2026a)
	3.82×10^{27}	0.01	...	2.40	...	Roth et al. (2026b)
	3.94×10^{27}	0.01	12/23/25	2.40	...	Cordiner et al. (2026)
CN	5.60×10^{24}	0.00	07/02/25	4.47	TTT	de la Fuente Marcos et al. (2025)
	5.60×10^{23}	0.00	07/04/25–07/05/25	4.41	X-SHOOTER	Alvarez-Candal et al. (2025)
	9.80×10^{23}	0.00	07/12/25	4.14	SNIFS	Hoogendam et al. (2025b)
	9.77×10^{23}	0.00	07/23/25	3.78	X-SHOOTER	Rahatgaonkar et al. (2025)
	1.29×10^{24}	1.05	07/27/25	3.65	...	...
	1.38×10^{24}	1.05	07/30/25	3.55	...	...
	5.25×10^{24}	1.10	08/09/25	3.23	...	...
	5.80×10^{24}	1.90	08/10/25	3.16	MDM	Salazar Manzano et al. (2025)
	1.70×10^{24}	1.99	08/12/25	3.14	UVES	Rahatgaonkar et al. (2025)
	4.82×10^{24}	0.34	...	3.14	MDM	Salazar Manzano et al. (2025)
	1.58×10^{24}	1.99	...	3.14	UVES	Hutsemékers et al. (2026b)
	3.80×10^{24}	1.02	08/14/25	3.07	X-SHOOTER	Rahatgaonkar et al. (2025)
	3.89×10^{24}	1.12	08/15/25	3.04	UVES	Rahatgaonkar et al. (2025)
	3.89×10^{24}	1.12	...	3.04	X-SHOOTER	Hutsemékers et al. (2026b)
	4.17×10^{24}	1.02	08/16/25	3.01	...	Rahatgaonkar et al. (2025)
	6.34×10^{24}	0.29	08/16/25	2.97	MDM	Salazar Manzano et al. (2025)
	7.17×10^{24}	0.38	08/17/25	2.94	...	...
	8.40×10^{23}	3.20	08/18/25	2.94	SNIFS	Hoogendam et al. (2025b)
	6.46×10^{24}	1.02	08/21/25	2.85	X-SHOOTER	Rahatgaonkar et al. (2025)
	1.70×10^{24}	0.50	08/24/25	2.75	KCWI	Hoogendam et al. (2025a)
	5.70×10^{24}	1.50	08/27/25	2.66	LRIS	Medler et al. (2026)
	6.16×10^{24}	1.10	08/28/25	2.64	UVES	Hutsemékers et al. (2026b)
	2.45×10^{25}	0.00	...	2.45	VLT/ESPRESSO	Luco et al. (2026)
	9.12×10^{24}	1.07	09/3/25–09/4/25	2.44	UVES	Hutsemékers et al. (2026b)
	3.55×10^{25}	0.00	09/03/25	2.42	VLT/ESPRESSO	Luco et al. (2026)
	8.91×10^{25}	0.00	09/08/25	2.28	...	Luco et al. (2026)
	4.07×10^{25}	1.51	09/09/25	2.25	...	Luco et al. (2026)
	1.07×10^{25}	1.02	09/10/25	2.25	UVES	Hutsemékers et al. (2026b)

	4.07×10^{25}	1.20	...	2.22	VLT/ESPRESSO	Luco et al. (2026)
	2.19×10^{25}	1.02	09/11/25	2.22	UVES	Hutsemékers et al. (2026b)
	3.24×10^{25}	1.38	...	2.19	VLT/ESPRESSO	Luco et al. (2026)
	2.88×10^{25}	1.02	09/12/25	2.19	UVES	Hutsemékers et al. (2026b)
	6.17×10^{25}	1.51	...	2.17	VLT/ESPRESSO	Luco et al. (2026)
	2.69×10^{25}	1.02	09/14/25	2.14	UVES	Hutsemékers et al. (2026b)
	1.17×10^{26}	1.17	09/15/25	2.08	VLT/ESPRESSO	Luco et al. (2026)
	1.41×10^{26}	1.41	09/16/25	2.06	...	...
	6.31×10^{25}	1.17	09/17/25	3.03	...	...
	6.03×10^{25}	1.15	09/18/25	2.00	...	...
	3.63×10^{25}	1.17	09/19/25	1.98	...	...
	4.07×10^{25}	1.05	09/20/25	1.97	X-SHOOTER	Hutsemékers et al. (2026b)
	7.76×10^{25}	1.15	...	1.95	VLT/ESPRESSO	Luco et al. (2026)
	5.01×10^{25}	1.05	09/22/25	1.92	X-SHOOTER	Hutsemékers et al. (2026b)
	4.90×10^{25}	1.20	...	1.90	VLT/ESPRESSO	Luco et al. (2026)
	5.25×10^{25}	1.05	09/23/25	1.90	X-SHOOTER	Hutsemékers et al. (2026b)
	7.94×10^{25}	1.20	...	1.87	VLT/ESPRESSO	Luco et al. (2026)
	5.37×10^{25}	1.05	09/24/25	1.87	X-SHOOTER	Hutsemékers et al. (2026b)
	1.55×10^{25}	1.15	...	1.85	VLT/ESPRESSO	Luco et al. (2026)
	5.37×10^{25}	1.05	09/25/25	1.85	X-SHOOTER	Hutsemékers et al. (2026b)
	1.60×10^{26}	0.50	11/16/25	1.50	KCWI	Hoogendam et al. (2026)
	8.91×10^{25}	1.29	11/25/25	1.67	VLT/ESPRESSO	Luco et al. (2026)
	6.17×10^{25}	1.38	11/26/25	1.70	...	Luco et al. (2026)
	8.13×10^{25}	1.15	11/27/25	1.72	...	Luco et al. (2026)
	3.20×10^{25}	1.20	11/30/25	1.79	SNIFS	Medler et al. (2026)
	2.18×10^{25}	1.05	12/02/25	1.85	BFOSC	Zhao et al. (2026)
	6.76×10^{25}	1.38	12/04/25	1.88	VLT/ESPRESSO	Luco et al. (2026)
	3.31×10^{25}	1.05	...	1.89	YFOSC	Zhao et al. (2026)
	8.91×10^{25}	1.29	12/06/25	1.93	VLT/ESPRESSO	Luco et al. (2026)
	3.09×10^{25}	1.05	...	1.95	YFOSC	Zhao et al. (2026)
	3.16×10^{25}	1.05	12/07/25	1.97	...	...
	1.95×10^{25}	1.05	12/17/25	2.25	...	...
	2.75×10^{25}	1.07	12/19/25	2.31	...	...
	1.45×10^{25}	1.15	12/20/25	2.33	BFOSC	...
	3.39×10^{25}	0.00	12/21/25	2.35	VLT/ESPRESSO	Luco et al. (2026)
	3.10×10^{24}	0.00	12/23/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)
	1.26×10^{25}	1.15	12/23/25	2.42	BFOSC	Zhao et al. (2026)
	8.91×10^{24}	1.15	12/26/25	2.51	BFOSC	Zhao et al. (2026)
	8.91×10^{24}	1.05	12/28/25	2.57	YFOSC	...
	6.03×10^{24}	1.07	1/11/26	3.00	...	...
	4.37×10^{24}	1.07	1/20/26	3.29	...	...
HCN	1.70×10^{24}	0.00	07/16/25–07/17/25	4.01–3.97	JCMT	Hinkle et al. (2025)
	1.50×10^{25}	0.00	08/07/25	3.29	JCMT	Coulson et al. (2025)
	2.10×10^{25}	0.00	08/15/25	3.00	...	...
	1.70×10^{25}	0.00	08/22/25	2.8	...	...
	1.40×10^{25}	0.00	08/28/25	2.81	...	...
	1.10×10^{25}	0.00	09/03/25	2.63	...	...
	1.20×10^{25}	0.50	09/07/25	2.33	...	...
	5.00×10^{24}	1.00	09/12/25	2.17	ALMA /ACA	Roth et al. (2025)
	4.00×10^{25}	1.70	09/14/25	2.14	JCMT	Coulson et al. (2025)
	1.00×10^{25}	0.10	09/15/25	2.11	ALMA /ACA	Roth et al. (2025)
	4.10×10^{25}	0.20	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
	6.75×10^{24}	0.54	12/22/25	2.37	ALMA /ACA	Cordiner et al. (2026)
OH	8.20×10^{26}	0.00	07/04/25–05/26/25	4.41	X-SHOOTER	Alvarez-Candal et al. (2025)
	1.48×10^{26}	0.00	07/27/25	3.14	UVES	Rahatgaonkar et al. (2025)
	8.00×10^{27}	0.00	08/26/25–09/03/25	2.54	TMRT	Li et al. (2026)
	3.80×10^{26}	1.10	08/28/25	2.64	UVES	Hutsemékers et al. (2026b)
	6.16×10^{26}	1.05	09/03/25–09/04/25	2.44	...	...

	1.32×10^{28}	0.47	09/09/25	2.27	TMRT	Li et al. (2026)
	1.89×10^{28}	0.37	09/18/25–09/23/25	1.96	...	...
	1.74×10^{27}	1.35	09/10/25	2.25	UVES	Hutsemékers et al. (2026b)
	2.14×10^{27}	1.26	09/12/25	2.19	...	...
Ni	1.62×10^{22}	1.35	07/23/25	3.78	X-SHOOTER	Rahatgaonkar et al. (2025)
	1.29×10^{22}	1.62	...	3.78	...	Hutsemékers et al. (2026b)
	2.75×10^{22}	1.44	07/27/25	3.65	...	Rahatgaonkar et al. (2025)
	2.24×10^{22}	1.62	...	3.65	...	Hutsemékers et al. (2026b)
	2.69×10^{22}	1.14	07/30/25	3.55	...	...
	3.71×10^{22}	1.32	...	3.55	...	Rahatgaonkar et al. (2025)
	1.32×10^{23}	1.12	08/09/25	3.23	...	...
	6.76×10^{22}	1.05	...	3.23	...	Hutsemékers et al. (2026b)
	6.61×10^{22}	1.15	08/12/25	3.14	UVES	...
	6.60×10^{22}	1.15	...	3.14	...	Rahatgaonkar et al. (2025)
	1.23×10^{23}	1.17	08/14/25	3.07	X-SHOOTER	...
	6.76×10^{22}	1.17	...	3.07	...	Hutsemékers et al. (2026b)
	5.49×10^{22}	1.20	08/15/25	3.04	UVES	...
	5.49×10^{24}	1.20	...	3.04	...	Rahatgaonkar et al. (2025)
	1.41×10^{24}	1.20	08/16/25	3.01	X-SHOOTER	...
	8.13×10^{22}	1.15	...	3.01	...	Hutsemékers et al. (2026b)
	1.99×10^{22}	1.17	08/21/25	2.85	...	Rahatgaonkar et al. (2025)
	1.20×10^{22}	1.15	...	2.85	X-SHOOTER	Hutsemékers et al. (2026b)
	1.86×10^{23}	1.12	08/28/25	2.64	UVES	...
	3.23×10^{23}	1.09	09/03/25–09/04/25	2.44	...	...
	5.62×10^{23}	1.07	09/10/25	2.25	...	...
	6.30×10^{23}	1.07	09/12/25	2.19	...	...
	1.12×10^{24}	1.12	09/20/25	1.97	X-SHOOTER	...
	1.78×10^{24}	1.10	09/22/25	1.92	...	...
	1.99×10^{24}	1.10	09/23/25	1.90	...	...
	1.95×10^{24}	1.01	09/24/25	1.87	...	...
	2.04×10^{24}	1.01	09/25/25	1.85	...	...
	7.47×10^{24}	0.27	11/16/25	1.50	KCWI	Hoogendam et al. (2026)
	9.77×10^{24}	1.05	12/02/25	1.85	BFOSC	Zhao et al. (2026)
	8.71×10^{24}	1.23	12/04/25	1.89	YFOSC	...
	7.76×10^{24}	1.23	12/06/25	1.95	...	...
	9.12×10^{23}	1.07	...	1.93	UVES	Hutsemékers et al. (2026a)
	6.76×10^{24}	1.23	12/07/25	1.97	YFOSC	Zhao et al. (2026)
	9.33×10^{23}	1.10	12/10/25	2.04	UVES	Hutsemékers et al. (2026a)
	7.78×10^{23}	1.07	12/15/25	2.18	...	...
	7.76×10^{24}	1.14	12/16/25	2.20	JWST/MIRI	Belyakov et al. (2026)
	5.01×10^{24}	1.23	12/17/25	2.25	YFOSC	Zhao et al. (2026)
	3.63×10^{24}	1.23	12/19/25	2.31	...	...
	4.7×10^{24}	1.05	12/20/25	2.33	BFOSC	...
	6.16×10^{23}	1.10	12/21/25	2.35	UVES	Hutsemékers et al. (2026a)
	3.98×10^{24}	1.05	12/23/25	2.42	BFOSC	Zhao et al. (2026)
	3.39×10^{24}	1.05	12/26/25	2.51	...	...
	4.57×10^{23}	1.10	...	2.50	UVES	Hutsemékers et al. (2026a)
	2.88×10^{24}	1.23	12/28/25	2.57	YFOSC	Zhao et al. (2026)
	1.82×10^{23}	1.12	01/11/26	2.99	UVES	Hutsemékers et al. (2026a)
	1.95×10^{24}	1.23	...	3.00	YFOSC	Zhao et al. (2026)
	1.14×10^{23}	1.15	01/19/26	3.24	UVES	Hutsemékers et al. (2026a)
	1.51×10^{24}	1.23	01/20/26	3.29	YFOSC	Zhao et al. (2026)
	7.59×10^{22}	1.17	01/27/26	3.50	UVES	Hutsemékers et al. (2026a)
	2.75×10^{22}	1.26	02/07/26	3.85	...	...
Fe	6.76×10^{21}	0.00	08/12/25	3.14	UVES	Hutsemékers et al. (2026b)
	1.38×10^{22}	0.00	08/15/25	3.04	...	...
	1.00×10^{22}	1.20	08/28/25	2.64	...	...
	4.00×10^{22}	1.10	09/03/25–09/04/25	2.44	...	...

	1.15×10^{23}	1.07	09/10/25	2.25	...	...
	1.58×10^{23}	1.05	09/12/15	2.19	...	...
	4.78×10^{23}	1.07	09/20/25	1.97	X-SHOOTER	...
	8.51×10^{23}	1.07	09/22/25	1.92	...	...
	1.00×10^{24}	1.05	09/23/25	1.90	...	...
	1.15×10^{24}	1.07	09/24/25	1.87	...	...
	1.17×10^{24}	1.07	09/25/25	1.85	...	...
	6.30×10^{24}	0.10	11/16/25	1.50	KCWI	Hoogendam et al. (2026)
	2.04×10^{24}	1.05	12/02/25	1.85	BFOSC	Zhao et al. (2026)
	1.99×10^{24}	1.23	12/04/25	1.89	YFOSC	Zhao et al. (2026)
	2.09×10^{24}	1.10	12/04/25	1.88	UVES	Hutsemékers et al. (2026a)
	8.32×10^{23}	1.02	12/06/25	1.93	...	...
	1.74×10^{24}	1.23	12/06/25	1.95	YFOSC	Zhao et al. (2026)
	1.48×10^{24}	1.23	12/07/25	1.97	...	...
	8.51×10^{23}	1.02	12/10/25	2.04	UVES	Hutsemékers et al. (2026a)
	6.03×10^{23}	1.02	12/15/25	2.18	...	...
	5.62×10^{24}	1.23	12/17/25	2.25	YFOSC	Zhao et al. (2026)
	4.07×10^{24}	1.23	12/19/25	2.31	...	...
	4.17×10^{24}	1.07	12/20/25	2.33	BFOSC	...
	3.16×10^{23}	1.02	12/21/25	2.35	UVES	Hutsemékers et al. (2026a)
	2.45×10^{23}	1.07	12/23/25	2.42	BFOSC	Zhao et al. (2026)
	1.86×10^{23}	1.02	12/26/25	2.50	UVES	Hutsemékers et al. (2026a)
	1.74×10^{23}	1.23	12/28/25	2.57	YFOSC	...
	4.27×10^{22}	1.10	01/11/26	2.99	UVES	Hutsemékers et al. (2026a)
	2.29×10^{22}	1.17	01/19/26	3.24	...	...
	1.38×10^{22}	1.50	01/27/26	3.50	...	...
C ₂	2.47×10^{24}	0.82	08/10/25	3.16	MDM	Salazar Manzano et al. (2025)
	2.84×10^{24}	0.95	08/12/25	3.14	...	...
	1.43×10^{24}	0.48	08/16/25	2.97	...	...
	1.43×10^{24}	0.48	08/17/25	2.94	...	...
	3.98×10^{24}	1.58	08/28/25	2.64	UVES	Hutsemékers et al. (2026b)
	3.98×10^{24}	1.58	09/03/25–09/04/25	2.44	...	...
	3.71×10^{24}	1.05	09/10/25	2.25	...	...
	4.78×10^{24}	1.12	09/12/25	2.19	...	...
	1.74×10^{25}	1.12	09/22/25	1.92	X-SHOOTER	...
	2.09×10^{25}	1.12	09/23/25	1.90	...	...
	2.57×10^{25}	1.12	09/24/25	1.87	...	...
	2.57×10^{25}	1.12	09/25/25	1.85	...	...
	8.80×10^{25}	0.80	11/16/25	1.50	KCWI	Hoogendam et al. (2026)
	6.46×10^{24}	1.05	12/02/25	1.85	BFOSC	Zhao et al. (2026)
	6.31×10^{24}	1.23	12/04/25	1.89	YFOSC	...
	5.75×10^{24}	1.23	12/06/25	1.95	...	...
	4.79×10^{24}	1.23	12/07/25	1.97	...	...
	2.00×10^{24}	1.23	12/17/25	2.25	...	...
	1.48×10^{24}	1.23	12/19/25	2.31	...	...
	1.48×10^{24}	1.07	12/20/25	2.33	BFOSC	...
	8.32×10^{23}	1.07	12/23/25	2.42	...	...
CH ₃ OH	5.00×10^{26}	0.60	08/28/25	2.64	ALMA /ACA	Roth et al. (2025)
	9.60×10^{26}	0.70	09/18/25	2.03	ALMA /ACA	Roth et al. (2025)
	1.21×10^{27}	0.05	09/22/25	1.92	ALMA /ACA	Roth et al. (2025)
	2.29×10^{27}	0.40	10/01/25	1.71	ALMA /ACA	Roth et al. (2025)
	3.38×10^{27}	0.05	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
	4.10×10^{27}	0.10	11/04/25	1.37	ALMA /ACA	Salazar Manzano et al. (2026)
	1.37×10^{26}	0.04	12/23/25	2.40	JWST/NIRSpec	Roth et al. (2026a)
¹³ CO ₂	2.65×10^{26}	0.03	...	2.40	...	Roth et al. (2026b)
	1.30×10^{25}	0.25	08/01/25–08/15/25	3.20	SPHEREx	Lisse et al. (2025a)
¹³ CO	9.40×10^{24}	0.00	12/23/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)
	2.49×10^{25}	0.00	12/23/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)

OCS	1.70×10^{24}	0.90	08/06/25	3.32	JWST	Cordiner et al. (2025)
	5.00×10^{26}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
	4.70×10^{24}	0.50	12/23/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)
C ₃	1.20×10^{24}	0.29	08/10/25	3.16	MDM	Salazar Manzano et al. (2025)
	8.90×10^{23}	3.00	08/12/25	3.14	...	...
	8.90×10^{23}	3.00	08/16/25	2.97	...	...
	9.40×10^{23}	3.10	08/17/25	2.94	...	...
	5.00×10^{24}	0.40	11/16/25	1.50	KCWI	Hoogendam et al. (2026)
	1.86×10^{24}	1.05	12/02/25	1.85	BFOSC	Zhao et al. (2026)
	2.29×10^{24}	1.05	12/04/25	1.89	YFOSC	...
	2.29×10^{24}	1.05	12/06/25	1.95	...	...
	2.04×10^{24}	1.05	12/07/25	1.97	...	...
	1.07×10^{24}	1.05	12/17/25	2.25	...	...
	1.62×10^{24}	1.07	12/19/25	2.31	...	...
	1.66×10^{24}	1.07	12/20/25	2.33	BFOSC	...
	1.95×10^{23}	1.07	12/23/25	2.42	...	...
misc. organics	2.00×10^{27}	0.50	12/08/25–12/15/25	2.0–2.20	SPHEREx	Lisse et al. (2026)
CH ₄	4.20×10^{26}	0.20	12/16/25	2.20	JWST/MIRI	Belyakov et al. (2026)
	1.90×10^{26}	0.02	12/23/25	2.40	JWST/NIRSpec	Roth et al. (2026a)
	2.06×10^{27}	0.01	...	2.40	...	Roth et al. (2026b)
	2.30×10^{26}	0.10	12/27/25	2.54	JWST/MIRI	Belyakov et al. (2026)
CH	3.24×10^{25}	1.07	12/02/25	1.85	BFOSC	Zhao et al. (2026)
	4.36×10^{25}	1.07	12/04/25	1.89	YFOSC	...
	4.07×10^{25}	1.07	12/06/25	1.95	...	...
	5.25×10^{25}	1.07	12/07/25	1.97	...	...
	2.34×10^{25}	1.10	12/17/25	2.25	...	...
	2.69×10^{25}	1.20	12/19/25	2.31	...	...
HDO	1.21×10^{27}	0.00	11/01/25–11/3/25	1.36–1.37	IRAM	Biver et al. (2026)
	3.20×10^{25}	0.19	12/23/25	2.40	JWST/NIRSpec	Cordiner et al. (2026)
CH ₃ CN	1.50×10^{25}	0.40	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
H ₂ CO	3.50×10^{26}	0.40	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
	3.90×10^{24}	0.90	12/23/15	2.40	JWST/NIRSpec	Roth et al. (2026a)
H ₂ S	3.90×10^{26}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
CS	1.70×10^{25}	0.40	11/01/25–11/30/25	1.36–1.37	IRAM	Biver et al. (2026)
HC ₃ N	2.10×10^{25}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
CH ₃ CHO	1.40×10^{26}	0.70	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
(CH ₃ OH) ₂	3.30×10^{26}	1.30	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
SO	2.70×10^{26}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
HCNO	2.50×10^{26}	1.10	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
HCOOH	1.09×10^{27}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
NH ₂ CHO	4.80×10^{25}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
CH ₃ OD	1.79×10^{26}	0.00	11/01/25–11/03/25	1.36–1.37	IRAM	Biver et al. (2026)
CH ₃ D	2.50×10^{25}	0.30	12/23/25	2.40	JWST/NIRSpec	Roth et al. (2026a)
C ₂ H ₆	2.00×10^{25}	0.20	12/23/25	2.40	JWST/NIRSpec	Roth et al. (2026a)

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